



Module	SE101.3 -OOP with Java – 25.3
Lab sheet Number	06
Areas covered	Inheritance

Question No 01

Create a class named Person with:

- name
- age

Include:

- Constructor
- Getter methods
- Setter methods

Create a subclass named Student with:

- course
- Getter and Setter

Create one Student object.

Use setter methods to assign values and display all details.

Question 02 - Library Management System

Create a parent class named Book with:

- title
- author
- isbn

Create a subclass named EBook with:

- fileSize
- fileFormat

Display all details of an EBook.

Question 03 – Cybersecurity User Management System

A company maintains information about users who access its network.

Create a superclass named User with:

- username
- email

Create a subclass named SecurityAnalyst with:

- assignedTool
- securityLevel

Display all details.

Example Tools:

Wireshark

Nmap

Burp Suite

Splunk

Question 04 – Network Infrastructure Monitoring System

Create a superclass named NetworkDevice with:

- deviceName
- ipAddress

Create a subclass named Firewall with:

- securityPolicy
- vendor

Display all details.

Example Vendors:

Cisco

Fortinet

Palo Alto

Question 05 - Banking System

Create a superclass named Account with:

- accountNumber
- balance

Create a subclass named SavingsAccount with:

- interestRate

Requirements:

- Use constructors to initialize all attributes.
- Create a method to calculate yearly interest
- Display all account details and the calculated interest

Question 06 – Incident Tracking System

Create a superclass named Incident with:

- incidentId
- incidentDate

Create a subclass named CyberAttack with:

- attackType
- severityLevel

Display all details.

Example Attack Types:

Phishing

Malware

Ransomware

DDoS

Question 07 – Secure Access Control System (Challenge Question)

An organization manages employee access to critical systems.

Create a superclass named Employee with:

- employeeId
- employeeName
- email

Create a subclass named CyberSecurityOfficer with:

- clearanceLevel
- monitoredSystems

Requirements:

- Use constructors to initialize all attributes.
- Use getter methods to display details.
- Create at least two CyberSecurityOfficer objects.

Display all information.